Cleaning or Recalling? Disentangling Memorization and World Knowledge in LLM-based Data Cleaning [Experiment, Analysis & Benchmark]

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ABSTRACT

High scores on public data-cleaning benchmarks can reflect knowledge of familiar entities, leaving unclear how well an LLM can repair unfamiliar data. We present CoRe, a controlled evaluation protocol linking recovery evidence, assigned values and repair credit. It compares benchmarks with fresh real records and name surrogates built from English words or syllables. The paired surrogates retain error positions and transfer error mechanisms, linking performance changes to the evidence each error leaves available for repair. Across 21 controlled tables and eight LLMs, we find large knowledge effects on Hospital's identity columns and on Movies for commercial models. On Movies, word renaming lowers DeepSeek-V4-Pro's year accuracy from 91% to 18% despite unchanged candidates. Knowledge effects persist on fresh records, while a continuation probe finds no positive evidence of benchmark-row memorization in the tested models. Yet recovering a familiar fact is not always the task's goal: models replace assigned movie directors with real-world credits, sometimes even under table-evidence instructions. Whether an edit receives repair credit then depends on the reference. On Rayyan, checks against title-matched PubMed records identify reference discrepancies, and 94% of TabClean's credited repairs reproduce the reference date transformation. These results connect the evidence supporting a repair to the task's preservation requirements and the reference that awards credit.

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1 INTRODUCTION

Data cleaning methods detect and repair erroneous values and are commonly evaluated on a small set of public benchmark tables [2, 10, 22, 51]. Large language models (LLMs) have opened new approaches to data management [17], including few-shot error detection and imputation competitive with specialized systems [37]. Recent cleaners use LLMs to edit values, derive rules or write cleaning programs [41, 42, 63, 71]. Their benchmark scores, however, can draw on information beyond the table being cleaned.

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Public cleaning benchmarks contain familiar hospitals, films, breweries and articles. Pretrained models can retrieve entity facts from their parameters [34, 47], while public tabular datasets can carry signals of prior exposure [6, 57]. A correct repair can therefore combine benchmark-row memory, knowledge of the entity (*entity knowledge*), knowledge of English word forms (*spelling knowledge*), and evidence within the table. Scores on familiar entities need not predict repair quality on unfamiliar data.

Example 1.1. Movies contains two distinct recovery problems. A Rotten Tomatoes slug can stand in place of an IMDb identifier. For *American Ultra*, recovering `tt3316948` from `american_ultra` requires the film's mapping; a fictitious slug does not determine its random identifier. Year ranges, however, retain the same candidates after renaming. DeepSeek-V4-Pro's repairs decline in both cases (Figure 1), although the year candidates remain available. The score changes therefore call for different explanations.

Controlled tabular evaluations already test recognition and completion of dataset instances (KnowTable [57]) and adherence to counterfactual database values (ContraTable [62]). Cleaning poses a further question: which input values should change, and what makes a change count as a repair? Interpreting scores requires connecting recovery evidence with the values to preserve and the reference used for scoring.

In this paper, we present CoRe, a protocol that compares repairs on dirty/reference pairs under controlled entity replacements. Each benchmark is compared with *fresh* real records, a *words surrogate* with fictitious English-word names, and a *syllable surrogate* with meaningless names. The original and its surrogates share rows and error positions, while fresh records carry the same kinds of errors. Paired substitutions test entity familiarity and word form. Fresh records test performance beyond the benchmark, while a separate continuation probe tests for benchmark-row memory.

The constructions retain the original cleaning problems by transferring error mechanisms, rather than adding arbitrary noise. This is particularly important for Movies, where errors often reflect differences between two legitimate sources. Fictitious names must also hide the entity without themselves looking erroneous. We build 21 controlled tables: nine each for Hospital and Movies across three seeds, and three for Beers. Their surrogates contain about 41,000 fictitious values, assessed by 506 acceptance checks. We evaluate four commercial and four open-weight LLMs, alongside Baran [32] and the program-generating cleaner TabClean [63], using about 23,000 prompts. Our main findings are as follows.

Familiar entities affect both recovery from missing mappings and the use of retained evidence. On Movies, DeepSeek-V4-Pro's year accuracy falls from 91% on real films to 18% on word names and

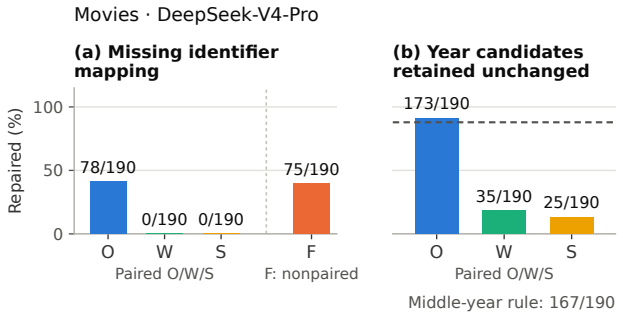


Figure 1: Recovery evidence in Movies. DeepSeek-V4-Pro repairs fewer identifiers when mappings disappear, but also fewer years when candidates remain. Labels give repaired/error cells; the middle-year rule matches 167/190 references. O/W/S are paired; F uses fresh records.

13% on syllable names, even though the candidates are unchanged (Figure 1). Across the four commercial models, Movies' identity-column fresh–words gaps are +0.17 to +0.42 at construction seed 0 (Table 7). On Hospital, three commercial models lose 0.59 to 0.72 repair F1 after syllable replacement on *identity columns*, which name or locate an entity.

The continuation probe finds no positive evidence of benchmark-row memorization in its tested models (Section 6). The knowledge effects also persist across construction seeds and, for DeepSeek-V4-Pro, with labeled examples and error positions. Within Qwen2.5, Hospital's total effect grows across the three model sizes. On Hospital and Beers, whole-table Baran and TabClean retain high scores under renaming, with repeated records and formatting patterns available for repair (Sections 7–8). The companion materials retain the three-dataset condition overview and its intervals.

Yet a familiar fact can be the wrong value to preserve. Commercial models rewrite 68%–100% of assigned directors of known films toward real-world credits. Table-evidence instructions reduce these rates unevenly, while hospital and brewery city rewrites are less frequent (Section 9).

The reference then determines repair credit. Rayyan contains systematic date and page conversions (Section 10). Stratified field checks against title-matched PubMed records yield a weighted reference-check failure rate of 95.2% (95% interval: 91.2–97.8%), using the audit input's 948-cell error-type composition. On TabClean's input version, 434 of 461 credited repairs (94%), after excluding development rows, reproduce the reference date transformation.
We make the following contributions:

- We propose CoRe, a controlled evaluation protocol that transfers benchmark errors to paired name surrogates and relates repair scores to the recovery evidence retained in the input (Section 3).
- We construct controlled tables for three benchmarks and provide construction and evaluation code, numerical witnesses and replay materials. Public statistical replay and authorized local validation provide distinct verification entries (Sections 4 and 11.2).
- We characterize both knowledge-dependent repair and directed rewriting of assigned values across eight LLMs, with separate continuation and information-condition checks (Sections 6–9).
- We identify systematic reference transformations in Rayyan and trace how they receive repair credit (Section 10).

Section 2 provides background and related work. Sections 3–5 describe the protocol, constructions and evaluation settings. Sections 6–10 present the analyses, followed by recommendations and limitations in Section 11.

2 BACKGROUND AND RELATED WORK

2.1 Cleaning Benchmarks

Cleaning systems draw on several kinds of evidence. Rules repair constraint violations or apply fixing patterns [11, 13, 61], while KATARA uses knowledge bases and crowdsourcing [12]. Learned detectors use a small set of labels [20, 33, 39]; Baran learns context models for repair [32], and HoloClean combines signals through probabilistic inference [52]. Experimental studies compare detection and repair tools [2, 40] and their effects on machine-learning models [1, 29]. ActiveClean integrates progressive cleaning with statistical model training [26]. BART controls detectability and repairability during error generation [3].

We select public dirty/reference pairs from Raha and Baran for their complementary error profiles (Table 1). Hospital's damaged names contrast with Movies' source-specific mappings and retained candidates. Beers adds a formatting contrast: its errors retain the information needed for repair. Rayyan's spreadsheet transformations instead expose what reference matching rewards. For cleaning we use 1,000-row tables. Movies and Beers retain all source-replaced and city-error rows, respectively, plus sampled benchmark rows. Flights and Tax serve only the continuation probe.

Identity columns name or locate an entity: provider number, name, address, city, ZIP code, county and phone in Hospital; identifier, title, director, writers and actors in Movies; and beer, brewery and city in Beers. We distinguish their recovery from repairs of other values, such as film years and formatting errors.

2.2 LLM-Based Cleaners

Few-shot prompting can match specialized systems on studied detection, imputation and matching tasks [37]. Subsequent work includes table-tuned and preprocessing models [28, 70] and entity matchers [46, 58]. LLM cleaners label errors (ZeroED [41]), iterate on repairs (IterClean [42]), derive ontological or semantic rules (LLMClean [4], Cocoon [71]), standardize data as agents (CleanAgent [48]), compile curation pipelines (SEED [9]), or generate cleaning programs (TabClean [63]). We study direct value editing and generated programs under controlled entity replacements.

2.3 Related Work

LLMs on enterprise data. Enterprise annotation and integration studies examine domain context, schemas, table width and sparsity [15, 25]. Bodensohn et al. also observe a bias toward familiar SQL despite explicit SIGNAL documentation [5]. CoRe compares repair and preservation under controlled entity replacements.

Memorization and contamination. LLMs can memorize training text, with stronger effects as models grow or data repeat [7, 8]. Detection methods include benchmark completion [18], likelihoods of canonical and shuffled orderings under exchangeability [45], and

Table 1: Benchmarks distributed with Raha and Baran. Parentheses give identity-column error cells. CMS: US Centers for Medicare & Medicaid Services.

Dataset	Entities	Source	Error cells	Kind of error	Use in this paper
Hospital	US hospitals	CMS registry	509 (219)	one character replaced by x	four conditions
Movies	films	IMDb and Rotten Tomatoes	1,965 (379)	the other source's value; formatting rules	four conditions
Beers	craft beers and breweries	CraftCans.com	565 (127)	state glued to city; alcohol % suffix	formatting control
Rayyan	biomedical articles	PubMed	948	spreadsheet conversions (Section 10)	quality analysis
Flights	flight times	38 web sources	not used for cleaning		memorization probe
Tax	personal tax records	—	not used for cleaning		memorization probe

least-likely token probabilities [56]. Benchmark-specific records and surveys document these issues [54, 67], while mitigation restricts exposure or refreshes questions [23, 64]. For tables, Bordt et al. [6] find strong memorization of Iris and Wine; Adult's signals concentrate in headers and initial rows. KnowTable [57] compares completion and existence queries over real, column-resampled, swapped and obfuscated variants, with non-neural baselines and statistical tests. Our comparison instead follows dirty/reference cell pairs: we transfer their error mechanisms to renamed entities and relate repair scores to the candidates and patterns left in the input. The continuation probe separately uses structural baselines to distinguish predictable cells from evidence of row memory.

Knowledge and counterfactual evaluation. Models store facts about entities, especially popular ones [24, 34, 47]. Entity substitutions in QA reveal conflicts between parametric and contextual answers [31, 66, 68], and counterfactual or low-probability variants expose dependence on familiar task instances [35, 36, 65]. DisentQA separates contextual and parametric answers [38]. In data management, ContraTable [62] pairs original and counterfactual databases to compare answer faithfulness across lookup, joins and temporal reasoning. A cleaner must both change errors and preserve assigned values. Our conflict test measures directed rewrites of reference-consistent cells toward familiar facts, alongside arbitrary edits (Section 9.2). Fictional ontologies and pseudowords also limit factual shortcuts in reasoning tasks [53, 55]. We contrast word and syllable names for cleaning.

Benchmark quality. Label errors can change model comparisons, particularly on mislabeled subsets and at higher error prevalence [43]. Spreadsheet conversion can also silently alter values [72]. Rayyan lets us trace credited repairs back to reference transformations and examine what the score rewards.

3 THE CORE PROTOCOL

CoRe connects the evidence for recovering values with the task's preservation requirements and repair credit (Figure 2). The four-condition comparison uses controlled dirty/reference pairs, holding the cleaner's prompt, information access and scoring fixed within a setting. Paired surrogates test dependence on familiar entities and word forms; fresh records test performance beyond the benchmark. Separate conflict and reference checks examine assigned-value preservation and the edits that receive credit (Sections 9 and 10).

3.1 Four Conditions and Their Comparisons

Besides the original (**O**), CoRe builds three tables with the same schema and size:

- **Fresh (F)** uses real records outside the benchmark: newer registry records for Hospital, or other films and beers. Their entities may remain familiar to a model.
- **Words surrogate (W)** replaces identity values one to one with fictitious names made of English words and random numeric identifiers. Non-identity values and error positions stay fixed.
- **Syllable surrogate (S)** uses syllable names in the same construction, retaining generic terms such as *hospital* and *street*.

Figure 3 illustrates naming and error transfer on Hospital rows.

Let s_O, s_F, s_W, s_S denote a method's repair F1 in a fixed column group. Table 2 defines three paired contrasts: O – W, W – S and O – S. They measure score changes under entity and word-form replacement; the first two sum to the third. Their interpretation uses the error profiles and method controls, since renaming changes candidate values and construction regularities as well as familiarity. We report F – W separately to examine knowledge effects on fresh records; it is not a component of O – S. O – F combines exposure and difficulty, so it supports the independent continuation probe rather than defining a pure memory effect.

3.2 What Makes an Error Recoverable?

Recovery evidence determines how to interpret the condition contrasts. Table 3 distinguishes *locally reversible* errors, whose information supports a conversion under a specified target convention (R1); *constrained* errors, which retain candidates or regularities (R2); and *underdetermined* errors, which need evidence absent from the shown chunk (R3). A column can contain several cases, and additional examples or records can change a cell's recoverability.

Where rules or candidates remain available, paired contrasts test their use after renaming. When the required mapping is absent, lower scores instead expose the recovery evidence removed by construction. Labels or repeated records can supply additional evidence (Section 8.1). The S score measures residual repair after renaming, including local rules and construction regularities.

3.3 Chunking, Prompting and Scoring

The main input contains 20 rows from different entities, preventing copying from another record of the same hospital or brewery while retaining cross-entity patterns. The same entity grouping gives O/W/S identical chunks, 50 per table. A zero-shot prompt requests JSON corrections with row, column and value fields, leaving correct cells unchanged. Other information settings are defined in Section 5.

Let E be the cells whose dirty and reference strings differ, P the cells effectively edited after no-ops are removed, and $R \subseteq P \cap E$

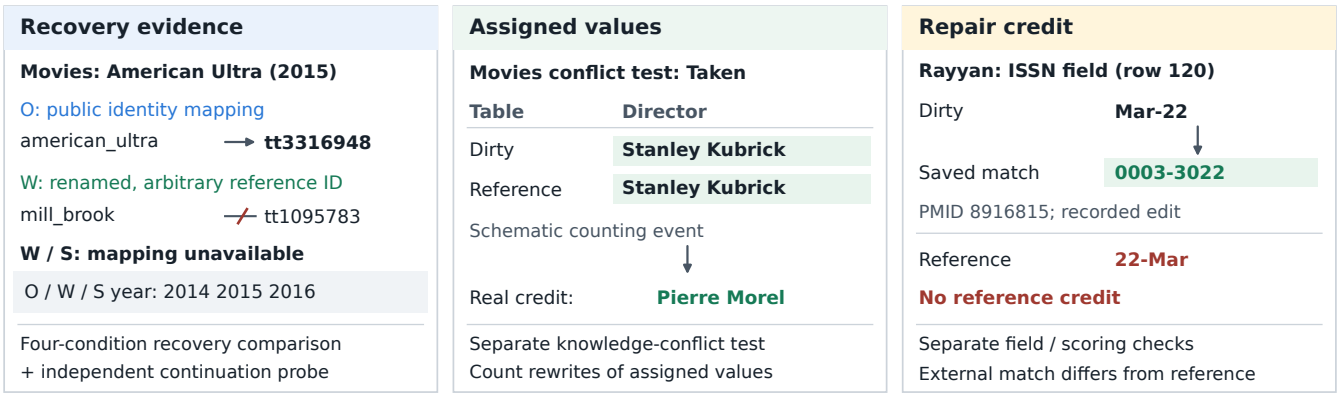


Figure 2: Three questions for interpreting repair scores: what evidence remains for recovery, which assigned values should be preserved, and which edits receive reference credit. The panels illustrate separate tests on their respective benchmark inputs.

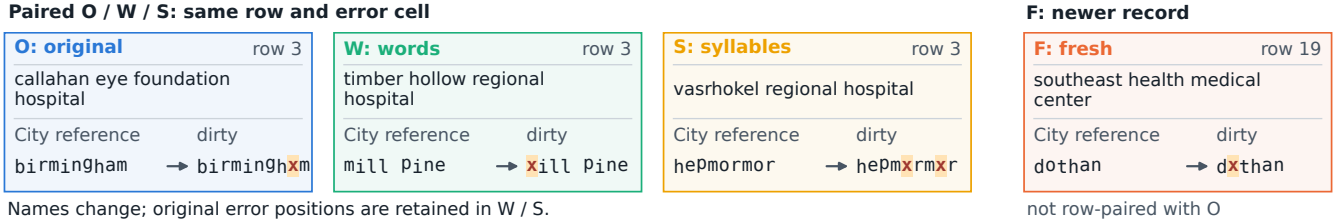


Figure 3: Hospital construction example. O/W/S share a row and city error cell while changing identity values. F uses a newer registry row with a newly drawn error location. The companion materials retain the complete identity-field example.

Table 2: Condition contrasts and their interpretation. O/W/S pair rows, error cells, chunks and non-identity values; F shares schema, size and error recipes or rules.

Contrast	Interpretation and comparison assumption
O−W	Paired entity-familiarity contrast; its magnitude also depends on the change from real names to English-word combinations.
W−S	Spelling contrast; interpretation depends on the difficulty of the two naming styles and any regularities they introduce.
O−S	Total knowledge effect under entity replacement, combining changes in entity familiarity and word form.
F−W	Fresh–words comparison; entity familiarity changes with record difficulty and, where recipes cannot transfer exactly, error composition.
O−F	Original–fresh comparison; benchmark exposure and table difficulty both change. Memorization is tested separately.

Table 3: Recovery evidence available in the shown input, under the cases defined in Section 3.2.

Case	Examples in the benchmark conditions
R1: locally reversible	Beers city/state split and percentage suffix; Movies thousands separators and integer ratings with an added `.0`.
R2: constrained	Movies year ranges; Hospital name typos with a matching candidate; syllable names with a restricted construction vocabulary.
R3: underdetermined	Random fictitious identifiers; omitted film credits; source-specific ratings and durations not recoverable from the other source.

those set to the reference value. Then

$$prec = \frac{|R|}{|P|}, \qquad rec = \frac{|R|}{|E|}, \qquad F1 = \frac{2 \cdot prec \cdot rec}{prec + rec},$$

and detection F1 uses $P \cap E$ in place of R. Edit and repaired-value comparisons ignore case and normalize whitespace, while E uses original strings. Thus Movies' five case-only title errors remain in E but cannot earn repair credit. Identity columns are reported separately from the others. The companion materials specify normalization, reply handling and the construction recipes used below.

4 BUILDING THE CONTROLLED TABLES

The controlled tables retain the benchmark's error mechanisms while varying entities and word forms. Error generators can vary detectability and repairability [3]; here, we start from the observed error profile and transfer it to renamed values or fresh records. Surrogates preserve non-identity cells and error positions, whereas fresh error counts depend on whether recipes and rules apply to the new records. Naming rules keep fictitious values from revealing identities or looking malformed. Table 4 summarizes the constructions, with full recipes and checks in the companion materials.

Table 4: Controlled-table sources and scale. Error counts in parentheses are identity cells; values renamed are distinct one-to-one replacements. Seeds cover F, W and S.

	Original (O)		Fresh (F)			W and S	
Dataset	Entities	Errors	Source	Entities	Errors	Values renamed	Seeds
Hospital	45 hospitals	509 (219)	CMS registry, July 2026 release	34 hospitals (33 also in O)	509 (219)	296	3
Movies	1,000 films	1,965 (379)	IMDb datasets, Sept. 2026	1,000 films	2,195 (371)	6,018	3
Beers	421 breweries	565 (127)	Wikidata, Sept. 2026	94 breweries	565 (127)	1,712	1

4.1 Hospital

Hospital's characteristic corruption replaces a character by `x`: 508 of 509 error cells follow this form, usually replacing every occurrence of that character. F uses the July 2026 CMS registry in the benchmark schema. Its 34 hospitals include 33 from O, but all quality measures have changed, giving new rows about largely the same entities. We retain the original per-column error counts and position-weighted substitution rule, with newly drawn error locations.

Surrogates rename each distinct identity value consistently and transfer the original errors by position (Figure 3). S uses an inventory of 24 syllables; W uses English-word combinations. Generic suffixes preserve column conventions, and numeric identifiers become random digits. The restricted syllable inventory supplies a recoverable regularity examined in Section 8.3.

4.2 Movies

Movies joins IMDb and Rotten Tomatoes records. Its 190 source-replaced rows, together with two formatting rules, produce the errors in Table 5. A film can have a Rotten Tomatoes slug instead of an IMDb identifier, a year range instead of a year, or credits that add or omit people. For example, *American Ultra* has `american_ultra` in place of `tt3316948`. Such differences require recovering the benchmark's preferred source value, rather than merely fixing spelling.

F matches each original film to another feature film in the September 2026 IMDb snapshot, using the same release year (or within two years) and the nearest logarithmic vote count among eligible films with at least 1,000 votes. The corresponding 190 rows receive source-style errors. People and title errors are transferred as recipes over names and edits; genre errors retain the other source's vocabulary. A recipe that cannot apply to its assigned cell is moved to another source-style row able to take it.

Some recipes remain inapplicable: four director and four writer errors stay clean, leaving 371 identity errors in F versus 379 in O/W/S. Formatting rules also change counts: the separator rule affects all 1,000 fresh rating counts, compared with 768 in O. F therefore has 2,195 errors at seed 0 versus 1,965 in O/W/S. These are fresh-record comparisons under transferred mechanisms, with the non-paired interpretation defined in Table 2.

Surrogates rename the 6,018 distinct identity values one to one. Titles require special care: retaining a real title's function words, numbers and punctuation can reveal the film. We generate title shapes independently at the original's aggregate rates, exclude real title words from W, and require generated titles to support their transferred edit recipes. Fictitious IMDb identifiers and slugs are then formed consistently with those titles.

Recovery evidence varies even within a column. Of 105 duration errors expressed in hours and minutes, unit conversion agrees with the reference in 50 and disagrees in 55. A locally available conversion can therefore differ from the preferred source value. Year ranges instead retain useful candidates, whose use we examine in Section 7.

4.3 Beers

All Beers errors retain the information needed for formatting repair. They append a percentage sign to a fractional alcohol value, or glue a state to a city and empty the state field. Removing the suffix or splitting the location restores the benchmark's convention. F uses 1,000 beers from 94 Wikidata breweries [59], excluding original beers and transferring the original error forms and counts.

Surrogates preserve generic terms and consistent relationships between brewery and city names. A one-word city stays one word, since a two-word fictitious city could resemble the city/state error itself. These choices keep the naming transformation from introducing the formatting problem the model is asked to solve.

4.4 Acceptance Checks and Construction Seeds

Acceptance checks cover structure, error counts and strength, naming leakage, fresh-source requirements and deterministic rebuilding. The tables pass 505 of 506 checks. Hospital W at seed 0 has fewer errors replacing three or more characters: 6.4%, versus 16.9% in O. The 10.5-point difference exceeds the eight-point tolerance by 2.5 points; error positions are unchanged. Full distributions and thresholds appear in the companion materials.

Naming and leakage account for 263 checks: one-to-one replacements, exclusion of real values, and title-shape comparisons against chance-level matching. Deterministic rebuilding checks that the same rules yield identical tables.

Hospital and Movies have three construction seeds: surrogates redraw names and applicable errors; F redraws errors on fixed clean records. Seed ranges capture construction variation, whereas bootstrap intervals summarize chunks of one construction. Main tables report seed 0. Hospital W at seeds 1 and 2 passes the strength check, with positive F − W and W − S contrasts for all eight cleaners. Table 12 gives construction ranges; the companion materials detail the passing-seed comparison.

5 EXPERIMENTAL SETUP

Models. We evaluate DeepSeek-V4-Pro [14], Qwen3.8-Max [50], GLM-5.3 [69] and GPT-6.1-Sol [44] as commercial cleaners. Locally, vLLM [27] uses greedy decoding for Qwen2.5 Instruct at 7B, 14B and 32B (AWQ) [49], and Llama-3.1-8B-Instruct [19]. OLMo-2-7B instruction/base variants [60] are auxiliary continuation models with a public pretraining corpus. Compact labels abbreviate these names; Qwen-7B/14B/32B denote Qwen2.5. Eight of GLM's 500

Table 5: Movies' 1,965 errors and their transfer mechanisms. Source-style errors occupy 190 rows, with formatting rules also applied table-wide. Copied errors transfer cell recipes; drawn errors follow measured rates. Slugs are the film identifiers used in Rotten Tomatoes URLs.

Column	Errors	In the original	Fresh table	Surrogates
identifier	190	Rotten Tomatoes slug (135 plain, 28 numbered, 27 with the year)	slug of the fresh title, forms at the original's rates	slug of the dirty fictitious title, same form
year	190	range "Y−1 Y Y+1"	same range	the original's cell
title	17	12 edit recipes (capitalization, & for *and*, accent, punctuation, subtitle, article)	the same edits on a fresh title	the same edits on the fictitious title
director	43	115 names added, 1 dropped, 17 respelled	copied name by name	copied name by name
writers	129	123 names added, 51 dropped, 20 respelled; 5 cells emptied	copied name by name	copied name by name
duration	187	Rotten Tomatoes format (105) or empty (80)	drawn at the original's rates	the original's cell
rating	261	score pair (165) or empty (22); ".0" added (74)	drawn at the original's rates	the original's cell
rating count	768	thousands separators on every count ≥ 1,000	the same rule	the original's cell
genre	180	Rotten Tomatoes genre vocabulary	copied slot by slot	the original's cell

Movies replies remain incomplete after retries and yield no parsed corrections. The companion materials detail reasoning settings, budgets, usage and exclusion sensitivity.

Methods and information. The *aligned* setting adds 20 labeled dirty/clean tuples and error positions to the main chunks. Baran [32] receives the same information, with three seeded runs, and also runs on whole tables with perfect detection or Raha [33]. We disable Baran's Wikipedia-revision value models and Raha's knowledge-base detector. TabClean uses DeepSeek-V4-Pro on whole tables, with its 150 labeled development rows excluded from scoring. These settings distinguish information available within a chunk from evidence supplied by repeated records and labels.

Scale. The study includes about 23,000 prompts for cleaning, aligned comparisons, knowledge conflicts and continuation. Their breakdown and run specifications are in the companion materials. Table 1 gives the dataset roles; the Rayyan analysis uses the version-specific inputs defined in Section 10.

Statistics. Condition differences use a chunk-level bootstrap [16] with 2,000 resamples, paired for O/W/S, and 95% percentile intervals. Each table's differences form a family, with significance at Holm-corrected [21] $p < 0.05$. Near-zero scores also use repaired-cell counts and paired-chunk sign tests, corrected over models. Main tables show construction seed 0; seed ranges describe the variation defined in Section 4.4.

Conflict intervals resample entity means 2,000 times under the metric in Section 9.2, conditioning on the selected entities in fixed table contexts. Rayyan intervals use 5,000 within-stratum resamples of saved binary checks, with fixed type weights, conditioning on the selected title matches and field criteria (Section 10.2). Both report 95% percentile intervals.

6 TESTING BENCHMARK MEMORIZATION

The continuation probe covers DeepSeek-V4-Pro, the four open-weight cleaners and both OLMo-2 variants. It asks for the next row after a header and ten consecutive rows, and tests whether dirty context elicits a clean reference value. The companion materials detail trial counts and context settings.

Separating structure from row memory. Many next-row values are predictable from repeated values and fixed record order. On Hospital, naive scoring gives DeepSeek-V4-Pro 47% continuation accuracy on cells absent from the context. We instead score *hard* cells, whose values are absent from the context column and not guessed by a column-mode or most-frequent-successor baseline, both fitted leave-one-out. Accuracy then falls to 2.4%. The baselines control for structural predictability.

Table 6: Highest hard-cell continuation accuracy over datasets for each context. Columns distinguish clean/dirty context, dataset naming and error-row selection. Leakage counts clean reference values written from dirty context.

	Anonymous		Named		Error	Clean value
Model	clean	dirty	clean	dirty	rows	leaked
DeepSeek-V4-Pro	0.065	0.060	0.085	0.065	—	3 / 479
OLMo-2-7B-Instruct	0.027	0.019	—	—	0.030	0 / 187
OLMo-2-7B (base)	0.011	0.011	—	—	—	1 / 92
Qwen2.5-7B	0.040	0.028	0.040	0.046	0.029	0 / 449
Llama-3.1-8B	0.042	0.019	0.014	0.009	0.016	1 / 490
Qwen2.5-14B	0.027	0.061	0.041	0.049	0.030	0 / 532
Qwen2.5-32B	0.005	0.037	0.013	0.028	0.029	0 / 537

No tested model exceeds 0.085 on hard cells (Table 6). DeepSeek's highest rates come from Beers, where its outputs name real beers of the contextual brewery rather than the next beer in the table. OLMo-2 base also has low accuracy (at most 0.011); an infini-gram [30] search of its public corpus finds only Hospital fragments in a GitHub issue and tool documentation, and none of the other tables.

Clean-value leakage. Given dirty context, DeepSeek writes the clean value of only 3 of 479 hard erroneous cells. The four open instruction cleaners together do so for 1 of 2,008. Values absent from the input provide a distinct reference-leakage test.

Original versus fresh. All four commercial cleaners score lower on O than F on Hospital (−0.03 to −0.07) and Movies (−0.09 to −0.25; Table 7). Some open models have the opposite sign, including Llama's +0.10 on Hospital, without positive evidence from the continuation probe. As defined in Table 2, O − F also includes difficulty changes and serves as supporting evidence. We next examine knowledge effects on original and fresh records.

Finding 1. After accounting for structural predictability, the continuation probe finds no positive evidence of benchmark-row memorization in any of the tested models.

7 KNOWLEDGE INFLATES CLEANING SCORES

Table 7 reports the eight-model condition scores and contrasts. The error profiles in Section 3.2 distinguish missing entity mappings from candidates and rules retained after renaming.

7.1 Hospital: Typos in Names

Replacing Hospital's entity names with syllables reduces DeepSeek-V4-Pro's identity-column repair F1 from 0.88 to 0.16. Qwen3.8-Max and GLM-5.3 have similarly large losses, giving a 0.59–0.72 range for these three models. Both paired steps matter: their W − S contrasts are +0.33 to +0.37, in addition to positive O − W contrasts. F − W is also large (+0.28 to +0.45), so the advantage of familiar entities persists on fresh records. GPT-6.1-Sol's stronger S performance is explained in Section 8.3.

Detection remains stronger than repair (Table 8). For example, DeepSeek detects S identity errors with F1 0.73 but repairs them with 0.16, while its non-identity repair scores remain between 0.91 and 0.95 across conditions. The aggregate gap separates error localization from value recovery: name candidates may constrain typos, while identifiers can remain underdetermined. Open models show smaller total effects, from +0.20 to +0.52 at seed 0.

Finding 2. DeepSeek-V4-Pro, Qwen3.8-Max and GLM-5.3 lose 0.59–0.72 repair F1 on Hospital's identity columns under syllable replacement. Entity familiarity and familiar word forms both contribute, while detection remains stronger than repair.

7.2 Movies: Candidates and Mappings

Year ranges retain useful candidates. In the 190 source-style rows, the year is "Y−1 Y Y+1" and the middle candidate matches 167 references (88%). DeepSeek is right 91% on real films, but only 18% on W and 13% on S, where it mostly chooses the first year (Figure 4). The unchanged candidates retain a useful constraint after renaming, although the prompt does not state the rule. Lower repair reflects limited use of this evidence on unfamiliar names. The other commercial models show the same pattern: 171–176 correct years on O, versus 24–48 on surrogates. Duration and genre repairs also decline after renaming (Table 10).

Rating-count formatting moves in the opposite direction. Its dirty and reference values are identical across O/W/S, yet DeepSeek repairs 2 of 768 counts on O and 90 on S by removing commas. All 90 S repairs occur in seven chunks. The rule remains locally available, but its use varies across chunks and naming conditions.

Identifiers require entity mappings. All commercial models score at most 0.05 on Movies' surrogate identity columns. Their fresh–words gaps are +0.17 to +0.42, while the paired O − W contrasts are smaller (Table 7). DeepSeek replaces 78 of 190 real slugs with the correct IMDb identifier and none on surrogates (Table 10). A fictitious slug does not determine its random identifier, so the repairs on real films use mappings absent after renaming.

Without identifiers, F − W remains +0.26 to +0.33 across the four commercial models, whereas O − W is +0.04 to +0.15. Original people-list errors often contain legitimate Rotten Tomatoes credits, making the reference's source choice difficult even when the film is known. GPT-6.1-Sol rarely guesses identifiers on O (10 of 190); its O − W of +0.07 is not significant after correction, whereas its fresh–words gap is.

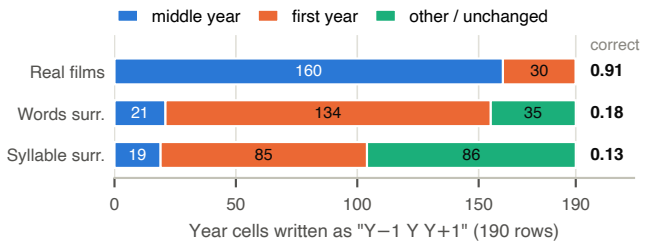


Figure 4: DeepSeek-V4-Pro's year repairs deteriorate after renaming, although all 190 year ranges are unchanged. Bars show the candidate written; the right panel shows repair accuracy. Taking the middle year matches 167 references (88%).

Open models repair at most 14 of 379 identity errors on O and three on W (Table 9). These near-zero results provide little evidence about their entity-knowledge effects on Movies.

GLM's O − W remains significant after excluding incomplete replies: +0.068 versus +0.070 on all chunks.

Finding 3. Movies' commercial-model fresh–words gaps are +0.17–+0.42 at seed 0, with surrogate identity repair F1 at most 0.05. Year repair also deteriorates despite retained candidate evidence.

7.3 Beers: Formatting Errors

Beers tests whether cleaners split city/state strings equally for real and fictitious names (Table 11). At seed 0, seven models show no significant positive O − S effect; their contrasts range from −0.08 to +0.04, with GPT's significant −0.03 in the opposite direction. Qwen3.8-Max instead repairs 122 of 127 cities on O, 108 on W and 82 on S, giving a significant total effect of +0.24. Its W-versus-S sign test has $p = 0.019$ uncorrected and 0.15 after correction over models. Dependence on familiar names therefore varies by model even for this formatting repair. F is descriptive because its brewery concentration and alcohol distribution differ from O (Section 4.3).

Finding 4. Qwen3.8-Max has a positive total knowledge effect of +0.24 on Beers' formatting errors; no significant positive effect appears in the other seven models.

7.4 Stability over Construction Seeds

Table 12 reports construction variation separately from chunk-bootstrap intervals. Hospital's total effect varies by at most 0.06 per model, and its spelling contrast by at most 0.05. F − W is more sensitive to the fresh error draw, with ranges reaching 0.11 for DeepSeek and 0.13 for open models. Movies' commercial-model gaps remain positive across seeds, including +0.39 to +0.44 for DeepSeek. The table gives the full model-specific ranges.

8 EVIDENCE AND KNOWLEDGE EFFECTS

8.1 Aligned Information and Repeated Records

The aligned setting adds labeled target-value examples and error positions without enlarging the main chunks. DeepSeek-V4-Pro improves across conditions but retains F − W gaps of +0.22 on Hospital and +0.58 on Movies (Figure 6). Hospital's rounded W and

Table 7: Identity-column repair F1 at seed 0. Nonpaired F−W uses fresh records; paired contrasts use O/W/S. †: Holm-corrected $p < 0.05$ over 40 differences (two-sided chunk bootstrap, 2,000 resamples).

	Hospital							Movies					
	Scores				Nonpaired	Paired		Scores				Nonpaired	Paired
Model	O	F	W	S	F−W	W−S	O−S	O	F	W	S	F−W	O−W
DeepSeek-V4-Pro	0.88	0.95	0.50	0.16	+0.45†	+0.34†	+0.72†	0.36	0.45	0.04	0.01	+0.41†	+0.32†
Qwen3.8-Max	0.90	0.93	0.55	0.21	+0.38†	+0.33†	+0.68†	0.30	0.44	0.02	0.00	+0.42†	+0.27†
GLM-5.3	0.78	0.83	0.55	0.19	+0.28†	+0.37†	+0.59†	0.11	0.20	0.04	0.01	+0.17†	+0.07†
GPT-6.1-Sol	0.94	0.98	0.71	0.62	+0.28†	+0.09†	+0.32†	0.12	0.37	0.05	0.04	+0.32†	+0.07
Qwen2.5-7B	0.22	0.23	0.07	0.02	+0.16†	+0.05	+0.20†	0.01	0.02	0.00	0.00	+0.02	+0.01
Llama-3.1-8B	0.31	0.21	0.02	0.00	+0.19†	+0.02	+0.30†	0.04	0.04	0.01	0.01	+0.04†	+0.03†
Qwen2.5-14B	0.44	0.42	0.19	0.04	+0.23†	+0.15†	+0.40†	0.06	0.04	0.01	0.01	+0.03	+0.05†
Qwen2.5-32B	0.55	0.48	0.31	0.03	+0.17†	+0.28†	+0.52†	0.03	0.03	0.00	0.00	+0.03	+0.03†

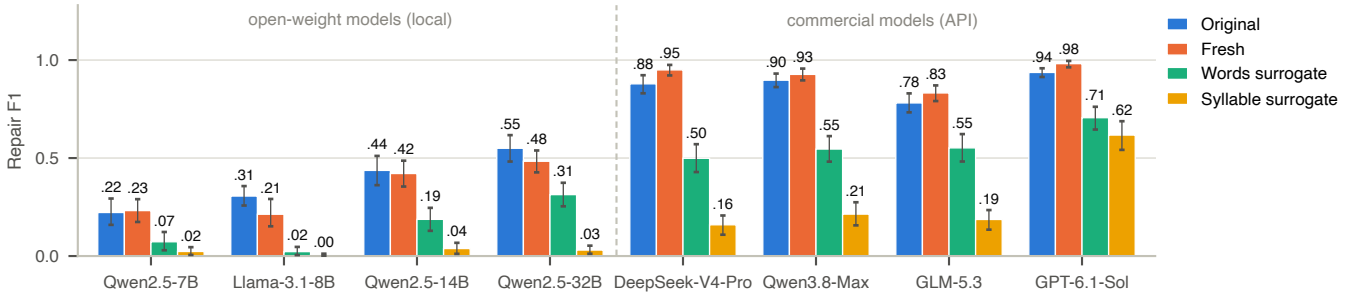


Figure 5: Hospital's total effect grows across Qwen2.5-7B/14B/32B, while S scores remain low. Identity-column repair F1, with Llama and commercial models for comparison; 95% bootstrap intervals.

Table 8: Detection and repair diverge on surrogate identity columns. Entries give detection F1 / repair F1; detection counts corrupted cells found regardless of the value proposed for that cell.

	Hospital			Movies		
Model	O	W	S	O	W	S
DeepSeek	1.0/.88	.95/.50	.73/.16	.56/.36	.41/.04	.47/.01
Qwen3.8	.98/.90	.97/.55	.72/.21	.56/.30	.57/.02	.52/.00
GLM-5.3	.98/.78	.97/.55	.72/.19	.26/.11	.12/.04	.08/.01
GPT-6.1	.95/.94	.71/.71	.65/.62	.17/.12	.11/.05	.09/.04

S scores differ by only 0.01. The added information largely closes their spelling contrast, while real entities retain an advantage.

Baran with the same chunks, labels and positions scores 0.27–0.32 on Hospital and zero on Movies identity columns. With whole-table perfect detection, it reaches 0.91–0.93 and 0.25–0.27, respectively, with little condition variation (Table 13). Whole-table access exposes repeated records that can supply values missing from the main chunks. DeepSeek still leads Baran in every aligned Hospital and Movies condition, but its advantage narrows on surrogates.

These comparisons expose two routes to improvement: labels and error positions improve DeepSeek's repair within a chunk, while repeated records supply values absent from that chunk. The former tests dependence on familiar entities under additional guidance; the latter changes the evidence from which a replacement can be recovered.

Table 9: Near-zero scores as repaired-cell counts. Hospital W/S have 219 identity errors; Movies O/W have 379. Chunk counts give more : fewer repairs in the first condition; sign-test p values are Holm-corrected over models.

	Hospital, W vs S			Movies, O vs W		
Model	W/S	Chunks	p	O/W	Chunks	p
DeepSeek	104/28	35:1	<.001	97/11	35:3	<.001
Qwen3.8	116/37	39:2	<.001	94/7	37:2	<.001
GLM-5.3	117/32	39:2	<.001	25/8	16:3	.022
GPT-6.1	120/100	19:6	.044	24/10	13:6	.334
Qwen-7B	11/4	8:4	.388	1/0	1:0	1
Llama-8B	5/1	4:0	.250	12/3	9:1	.086
Qwen-14B	35/7	23:2	<.001	14/3	10:0	.012
Qwen-32B	63/7	35:0	<.001	7/1	6:0	.094

8.2 Model Scale

Within Qwen2.5, Hospital's O − S grows from +0.20 at 7B to +0.40 at 14B and +0.52 at 32B, in the same order at every construction seed (Figure 5; Table 12). S repair F1 stays low at 0.02, 0.04 and 0.03. Higher original scores thus accompany a larger gap under entity replacement, with little improvement on unfamiliar names in this family and benchmark.

Table 10: Where knowledge acts on Movies: DeepSeek-V4-Pro repairs by column (repaired / errors). Identity columns are listed first; each condition uses its own error counts.

Column	O	F	W	S
identifier	78 / 190	75 / 190	0 / 190	0 / 190
title	0 / 17	4 / 17	1 / 17	0 / 17
director	18 / 43	27 / 39	7 / 43	2 / 43
writers	1 / 129	27 / 125	3 / 129	1 / 129
year	173 / 190	183 / 190	35 / 190	25 / 190
duration	90 / 187	101 / 187	42 / 187	41 / 187
genre	44 / 180	70 / 167	3 / 180	1 / 180
rating	30 / 261	73 / 280	9 / 261	9 / 261
rating count	2 / 768	26 / 1000	24 / 768	90 / 768

Table 11: Qwen3.8-Max shows a positive knowledge effect on Beers. Identity-column F1, total O–S, repaired cities (O/F/W/S, of 127), and W-versus-S chunk sign tests, Holm-corrected over models. *F is descriptive because its brewery concentration and alcohol distribution differ. †: significant as in Table 7, corrected over eight differences.

Model	O	F*	W	S	O−S	Cities repaired	p
DeepSeek-V4-Pro	0.27	0.24	0.11	0.28	−0.01	20 / 18 / 8 / 22	0.644
Qwen3.8-Max	0.94	0.61	0.82	0.69	+0.24†	122 / 67 / 108 / 82	0.152
GLM-5.3	0.94	0.88	0.89	0.94	+0.00	120 / 110 / 116 / 117	1.000
GPT-6.1-Sol	0.96	0.89	0.98	1.00	−0.03†	127 / 115 / 127 / 127	1.000
Qwen2.5-7B	0.00	0.10	0.04	0.05	−0.05	0 / 7 / 4 / 5	1.000
Llama-3.1-8B	0.18	0.27	0.11	0.14	+0.04	23 / 28 / 14 / 17	1.000
Qwen2.5-14B	0.16	0.05	0.11	0.19	−0.03	13 / 4 / 15 / 20	1.000
Qwen2.5-32B	0.03	0.00	0.04	0.11	−0.08	2 / 0 / 3 / 10	1.000

Table 12: Identity-column contrast ranges over three construction seeds; seed 0 is in Table 7. Fresh clean records are fixed while errors are redrawn.

	Hospital			Movies	
Model	F−W	W−S	O−S	F−W	O−W
DeepSeek	.34–.45	.33–.36	.66–.72	.39–.44	.31–.32
Qwen3.8	.34–.38	.33–.38	.68–.69	.38–.42	.26–.28
GLM-5.3	.28–.29	.34–.37	.57–.61	.14–.19	.07–.08
GPT-6.1	.28–.29	.09–.10	.32–.34	.32–.35	.07
Qwen-7B	.16–.22	.02–.07	.19–.20	.01–.03	.00–.01
Llama-8B	.14–.24	.01–.04	.30	.03–.04	.03
Qwen-14B	.23–.31	.12–.17	.37–.42	.03–.06	.05
Qwen-32B	.17–.30	.27–.28	.49–.52	.02–.03	.01–.03

8.3 Recoverable Syllable Names

GPT-6.1-Sol repairs 100 of Hospital S's 219 identity errors with 105 edits, compared with 28–37 repairs for the other commercial models. The constructor's 24-syllable inventory is represented in every chunk and permits a unique restoration for 95 corrupted names. GPT restores 74 of them; the other commercial models restore 8–12 and open models at most one (Table 14). The inventory is used by our analysis, not supplied to the cleaner. On digits, which it does not constrain, GPT repairs 6 of 97, like the others. Its stronger S performance is concentrated on names recoverable from the restricted vocabulary. Name-replacement gaps thus reflect a model's use of retained evidence as well as familiar entities.

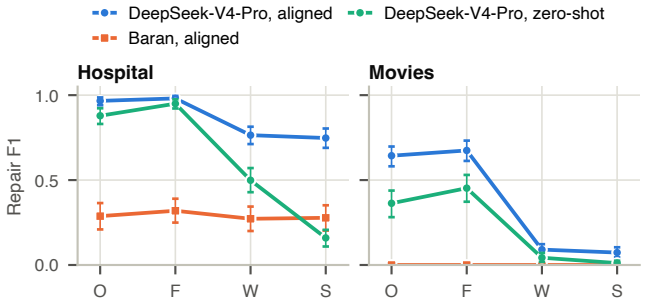


Figure 6: Labels and error positions improve DeepSeek-V4-Pro's repair but retain a fresh–words gap. Identity-column F1 in zero-shot and aligned settings; Baran is the aligned mean of three runs. Intervals are 95% chunk-bootstrap intervals.

Table 13: Identity-column repair F1 under aligned and whole-table information. Aligned uses the same chunks, 20 labeled tuples and error positions; Baran is the mean of three runs. Whole-table Baran uses perfect detection. TabClean uses DeepSeek-V4-Pro with development rows excluded.

Method	O	F	W	S
Hospital				
DeepSeek, aligned	0.97	0.98	0.76	0.75
Baran, aligned	0.29	0.32	0.27	0.28
Baran, whole table	0.91	0.93	0.93	0.93
TabClean	1.00	1.00	0.96	0.99
Movies				
DeepSeek, aligned	0.64	0.67	0.09	0.07
Baran, aligned	0.00	0.00	0.00	0.00
Baran, whole table	0.27	0.26	0.26	0.25
TabClean	0.00	0.00	0.03	0.00
Beers				
DeepSeek, aligned	1.00	1.00	1.00	1.00
Baran, aligned	0.19	0.77	0.20	0.22
Baran, whole table	0.94	1.00	0.94	0.96
TabClean	1.00	1.00	1.00	1.00

Table 14: Hospital S repairs by recovery category. Unique and Ambig. use the 24-syllable inventory; Generic covers English terms, and Digits covers numeric tokens. Headers give cell counts, covering 217 of 219 errors, with two address-spacing errors excluded from these counts.

Model	Unique (95)	Ambig. (11)	Generic (14)	Digits (97)
DeepSeek-V4-Pro	8	0	13	5
Qwen3.8-Max	12	0	12	11
GLM-5.3	10	0	12	8
GPT-6.1-Sol	74	4	14	6
Open models	0–1	0	0–3	0–6

8.4 Program-Generating Cleaner

TabClean [63] generates cleaning programs. Its identity-column scores stay near perfect on Hospital and Beers, but near zero on

Table 15: A conflict target with the same assigned Director reference in three name versions. The real credit of *Taken* (2008) is Pierre Morel; all other cells remain identical.

Version	Name	Year	Director
Exact name	Taken	2008	Stanley Kubrick
One-letter name	Takun	2008	Stanley Kubrick
Surrogate control	Drikel Joujou	2008	Stanley Kubrick

Movies (Table 13). For example, its Beers output splits "San Francisco CA" into city and state, both retained in the input. Movies' source mappings and omitted credits require external evidence.

Finding 5. Labels and error positions improve DeepSeek-V4-Pro's repair without closing its fresh–words gaps. Whole-table Baran and TabClean remain strong under renaming on Hospital and Beers, but weak on Movies.

9 KNOWLEDGE OVERRIDES THE TABLE

Movies often pairs legitimate values from different sources, so a familiar credit alone does not identify the task's target. We next fix a value to preserve and test whether a familiar name draws the cleaner toward the real-world fact.

9.1 Constructing Knowledge Conflicts

Starting from S, we assign target cells another real director or location in both dirty and reference tables. Correctness here means preserving the assigned target, although other cells in the row may still contain errors. Table 15 illustrates the assignment of Stanley Kubrick to *Taken*, whose real director is Pierre Morel.

Each target has an exact real name, a one-vowel-changed name and its syllable control. Only the name differs across versions; attributes, other values, error positions and chunks are fixed. Hospital and Beers receive internally consistent locations. Movies also updates related director occurrences consistently, preventing another column from supplying the old director as a candidate. The companion materials specify these cross-column replacements.

The common pool contains 57 films, 24 hospitals and 48 breweries known to DeepSeek-V4-Pro. Main chunks contain three targets and 17 other-entity fillers, with one row per target. Supplementary Hospital/Beers conditions use three rows per target. Every model receives the same pool, but rates use its own subset passing separate factual queries: 52–57 films, 22–24 hospitals and 7–48 breweries. Film knowledge accepts a credited full name or a surname of at least four letters; location knowledge requires matching city and state. These are conditional rates within the fixed pool.

The standard prompt asks for cleaning while preserving correct cells, without specifying source priority. The *table-evidence prompt* explicitly favors input-table evidence over real-world knowledge. Both use the same chunks and known subsets.

9.2 Measuring Rewrites

A directed rewrite changes the assigned Director or City to the paired entity's real-world fact. Director outputs must contain a full credited name; cities must match the normalized real city. The one-letter-name variant instead measures restoration of the exact real name. These events are scored separately from any change

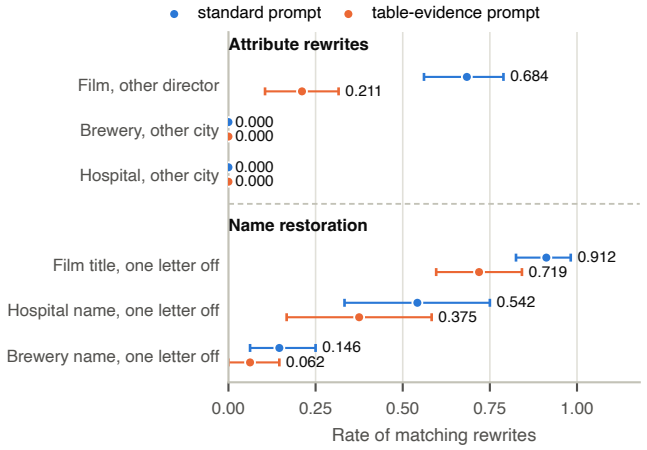


Figure 7: DeepSeek-V4-Pro rewrites assigned directors more often than cities. Attribute rewrites and name restoration are separate events, using 57 films, 24 hospitals or 48 breweries with one row per target and 95% entity-bootstrap intervals. Controls have zero directed rewrites.

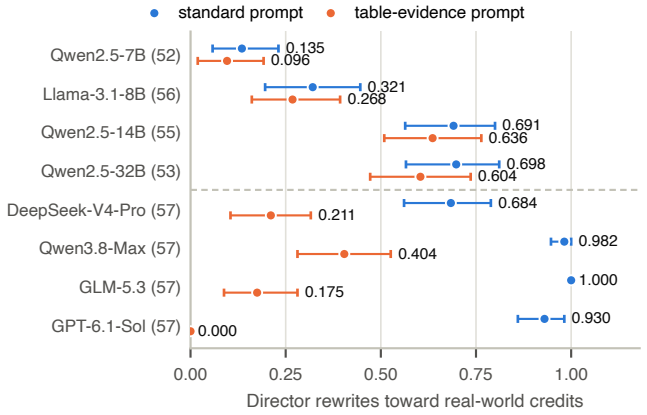


Figure 8: Table-evidence instructions reduce director rewrites unevenly. Rates divide matching real-credit rewrites by known films, with 95% entity-bootstrap intervals. Parentheses give denominators, unchanged between prompts; Table 16 gives counts and control edits.

to the target cell. Output parsing and matching normalization are specified in the companion materials.

Let E_m contain entities passing model m's factual queries, s be target rows per entity, and z_{ej} indicate a matching rewrite in row j of entity e. We report

$$\widehat{r}_m = \frac{1}{|E_m|} \sum_{e \in E_m} \frac{1}{s} \sum_{j=1}^{s} z_{ej}.$$

For Movies, $s = 1$, giving rewritten known films divided by known films. For $s = 3$, row indicators are first averaged within entities. Targets without a matching correction, including unusable replies, remain in the denominator. Surrogate controls use the same known subset and paired fact. Entity-bootstrap intervals follow Section 5.

Table 16: Director rewrite counts among known films (n/N). Exact-name targets count edits matching a real-world director credit. Surrogate controls count any Director change; none matches the paired film's real credit.

	Exact-name target		Surrogate control	
Model	Standard	Evidence	Standard	Evidence
DeepSeek-V4-Pro	39/57	12/57	2/57	0/57
Qwen3.8-Max	56/57	23/57	14/57	3/57
GLM-5.3	57/57	10/57	10/57	1/57
GPT-6.1-Sol	53/57	0/57	7/57	0/57
Qwen2.5-7B	7/52	5/52	0/52	0/52
Llama-3.1-8B	18/56	15/56	2/56	2/56
Qwen2.5-14B	38/55	35/55	2/55	0/55
Qwen2.5-32B	37/53	32/53	0/53	0/53

9.3 Results

Familiar names draw factual rewrites. Commercial models know the same 57 films and rewrite 68%–100% of their assigned directors under the standard prompt (Figure 8). DeepSeek, for example, rewrites 39/57. Controls also receive Director edits, but none matches the paired film's real credit (Table 16). The paired names separate directed factual rewrites from other edits.

Within Qwen2.5, conditional rates rise from 0.135 at 7B to 0.691 at 14B and 0.698 at 32B. Their known subsets differ, as shown in Table 16. Llama knows 56 films and rewrites 18 (0.321).

The effect depends on the attribute. Standard-prompt city rewrites reach at most 0.042 on Hospital and 0.140 on Beers across models. This is consistent with the distinction between a particular directing credit and an institution that may operate at multiple locations. DeepSeek also restores 91% of one-letter-changed film names (Figure 7), a possible spelling correction distinct from overriding an assigned attribute.

Source-priority instructions help unevenly. With the table-evidence prompt, GPT makes no directed director rewrite (0/57). DeepSeek still rewrites 12 of 57 directors; GLM and Qwen3.8-Max rewrite 10/57 and 23/57, respectively. Rates for Qwen2.5 at 14B and 32B remain above 0.60 (Table 16). These rewrites remain invalid under the prompt's explicit table priority.

Finding 6. For known films, commercial models rewrite 68%–100% of assigned directors toward real credits, while city rewrites are rare. Table-evidence instructions reduce their director rates to 0.000–0.404, but Qwen2.5-14B/32B retain rates above 0.60.

10 BENCHMARK QUALITY

The conflict test fixes values to preserve. Rayyan presents a complementary problem: a cleaner can recover a value supported by an external record and still receive no credit. Conversely, reproducing the reference's transformation can count as a repair.

10.1 Systematic Reference Transformations

Rayyan's audit pair has 1,000 rows and eleven columns, differing in 948 cells. Its values exhibit patterns characteristic of spreadsheet conversion [72]. All 722 date differences map input `m/d/yy` to `d/int(yy)/mm`, with a two-digit input month in the final position: for example, `1/1/13` becomes `1/13/01`. Similarly, 31 of 32 pagination differences turn page ranges into month-name strings, such as `70-6` to `Jun-70`.

Table 17: Rayyan field checks: population N_h, sample n_h, title matches m_h, failed reference and passed dirty checks per matched cell. A dash denotes an unassessed dirty check.

Check	N_h	n_h	m_h	Reference fail b_h/m_h	Dirty pass d_h/m_h
Year	722	80	44	43/44	42/44
Changed words	107	80	38	38/38	—
Vol./issue	75	75	46	28/46	17/46
Pages	32	32	18	18/18	17/18
ISSN	12	12	6	6/6	0/6

10.2 Stratified PubMed Checks

We sample up to 80 cells per error type, including all volume/issue, pagination and ISSN differences, for 279 cells. Automatic title matching selects the highest-similarity PubMed candidate among up to five results at a normalized similarity of at least 0.9. This yields 152 matched cells associated with 121 PMIDs: 54.5% overall and 47.5%–61.3% across types (Table 17). Query construction and retrieval provenance appear in the companion materials.

Field-specific checks. The date field `article_jcreated_at` is checked for agreement with PubMed's publication year; the dataset does not document its exact semantics. Volume and issue checks measure presence, using reference `-1` and dirty empty strings as absence. Two reference volumes of 1, for example, pass presence while differing numerically from saved PubMed values 31 and 21. Pagination matches the page string, and ISSN accepts print or electronic identifiers. For encoding differences, a nonempty set of reference-side changed words must all occur in the case-folded PubMed word set, with accents retained. Encoding's dirty side is unassessed.

Let N_h be the error-cell population of type h, m_h its title-matched count, and b_h the matched references failing that field's check. The weighted reference-check failure rate is

$$\widehat{q} = \sum_h \frac{N_h}{948} \frac{b_h}{m_h}. \tag{1}$$

Thus each within-type proportion uses matched cells, while weights follow the 948-cell audit composition. The result is 95.2% (95% interval: 91.2–97.8%; Table 17). Dates account for 722 of 948 error cells and most of the weight. The interval conditions on the saved title matches and field checks; matching uncertainty and dependence between cells sharing a PMID are outside this estimate.

10.3 What the Reference Rewards

On the audit input, DeepSeek-V4-Pro's zero-shot output scores F1 0.002. For example, it changes dirty ISSN `Mar-22` to `0003-3022`, matching the saved external ISSN, but the reference is `22-Mar`. Scoring rejects this external match because the reference differs.

TabClean's distributed Rayyan input has 960 difference cells, rather than the audit's 948. Both are 1,000-row, eleven-column tables with the same date-conversion pattern. We score the archived output against its actual input pair and exclude 150 development rows. The remaining 850 rows contain 808 errors; 498 effective edits

yield 461 credited repairs, giving $F_1 = 2 \times 461/(498 + 808) = 0.71$ after rounding. The companion materials record the input-version differences, development indices and saved output.

External checks reverse the date comparison: 42/44 matched dirty dates pass the year check, whereas 43/44 references fail. Yet 434 of TabClean's 461 credited repairs (94%) reproduce the reference date transformation. Reference matching therefore rewards this conversion while rejecting the externally matched ISSN.

10.4 Reference Issues in Other Benchmarks

Hospital's reference places Helen Keller Memorial Hospital in Jefferson County, although its city, Sheffield, is in Colbert County. Three commercial models correct some of these cells and are penalized as false edits. These smaller issues illustrate the same distinction between reference agreement and external correctness. The companion materials give their counts and an example of duplicate writer credits in Movies.

Finding 7. Rayyan's systematic date and page conversions affect repair credit. Stratified field checks against title-matched PubMed records give 95.2% weighted reference-check failure (95% interval: 91.2–97.8%), using the 948-cell type composition. On TabClean's version, 434/461 credited repairs (94%), after excluding 150 development rows, reproduce the reference date transformation.

11 LESSONS AND RECOMMENDATIONS

Interpret scores through recovery evidence. An identifier may lose its mapping after renaming, while a year range retains its candidates (Section 3.2). Report formatting repairs separately once the target convention is established: a local conversion and a source-specific value pose different recovery problems. Compare original, word and syllable conditions to examine dependence on familiar entities and word forms (Findings 2–4). Fresh records test transfer to new rows whose entities may remain familiar.

Align labels, error positions and supporting records. Chunk-based editing and whole-table repair offer different evidence. Compare aligned settings before attributing score gaps to method capability; repeated records can supply candidates absent from a chunk (Finding 5). Equal detection information need not supply equal evidence for recovering values.

Set source priority and test value preservation. Movies requires selecting a preferred source, whereas the conflict test requires retaining an assigned director. Count factual rewrites separately from other edits: a real credit can still be the wrong update for that task (Finding 6).

Check which edits receive repair credit. Rayyan credits date conversion but rejects an externally matched ISSN. Compare fields with independent records under explicit criteria, and bind scores to input versions and development exclusions (Finding 7).

11.1 Applying CoRe to Other Benchmarks

Adapting CoRe leaves the cleaner unchanged but requires an error profile, a compatible fresh source, two naming styles and acceptance checks. Profiling identifies the available recovery evidence and the intended target, including source priorities and reference transformations. The companion materials provide an application guide. Matching and imputation offer future settings.

11.2 Artifact and Result Replay

The statistical core supports offline replay of the condition contrasts, conflict rates and Rayyan checks. It provides Hospital/Beers tables and archived evidence; Movies uses numerical witnesses and saved continuation summaries. A separate authorized local entry checks historical movie inputs, replies and scoring-code hashes before re-scoring. The companion materials map C1–C3 to their input versions, executable entries and reported results, distinguishing public numerical replay from reader-supplied row-level validation.

11.3 Limitations

The datasets diagnose recovery mechanisms and reference transformations in entity-rich public tables, rather than estimate their prevalence across cleaning workloads. Transfer to private data also depends on size, sparsity, task complexity and the availability of supporting records. Our comparisons change name forms as well as familiarity, and F introduces record-difficulty differences (Table 2). Beers F has different brewery concentration and alcohol distributions, while Hospital's restricted syllables provide the regularity observed for GPT (Section 8.3). The study covers eight commercial and open-weight cleaners on English tables concerning mainly US entities, with decoding settings in the companion materials; it does not cover Claude or Gemini.

Intervals summarize fixed constructions and archived replies, rather than repeated API calls. Chunk resampling does not separately estimate dependence between chunks sharing an entity; entity resampling in the conflict test leaves shared-chunk dependence outside its uncertainty estimate. The controlled conflict measures source priority, not the prevalence of conflicting values in private data. Whole-table results show how additional records can supply evidence absent from the main chunks.

12 CONCLUSION

CoRe connects the evidence used to recover values with the targets that determine preservation and repair credit. Paired entity replacements and fresh records test recovery from familiar and unfamiliar inputs. Across eight LLMs, familiar entities contribute substantially to Hospital's identity-column scores and to Movies scores for commercial models. Movies illustrates why the retained evidence matters: DeepSeek-V4-Pro's repair falls both when identifier mappings disappear and when year candidates remain. The continuation probe finds no positive evidence of benchmark-row memorization in the tested models. Direct editing and table-based repair draw on different evidence, so information access also shapes the observed differences in repair.

Recovery alone does not determine whether an edit serves the task. Models can replace assigned directors with familiar real credits, with uneven reductions under table-evidence instructions. Repair credit ultimately depends on the reference. Rayyan's stratified field checks and credited date conversions show how a reference can reward its own transformations. Evaluations should therefore connect recovery on unfamiliar entities with source-priority and value-preservation tests, and inspect the targets used to score repairs. The construction principles provide a basis for studying these questions in other entity-based data preparation tasks.

REFERENCES

[1] Mohamed Abdelaal, Christian Hammacher, and Harald Schoening. 2023. REIN: A Comprehensive Benchmark Framework for Data Cleaning Methods in ML Pipelines. In *Proceedings of the 26th International Conference on Extending Database Technology (EDBT)*. 499–511. https://doi.org/10.48786/edbt.2023.43 arXiv:2302.04702.

[2] Ziawasch Abedjan, Xu Chu, Dong Deng, Raul Castro Fernandez, Ihab F. Ilyas, Mourad Ouzzani, Paolo Papotti, Michael Stonebraker, and Nan Tang. 2016. Detecting data errors: Where are we and what needs to be done? *Proceedings of the VLDB Endowment* 9, 12 (2016), 993–1004. https://doi.org/10.14778/2994509.2994518

[3] Patricia C. Arocena, Boris Glavic, Giansalvatore Mecca, Renée J. Miller, Paolo Papotti, and Donatello Santoro. 2015. Messing up with BART: Error generation for evaluating data-cleaning algorithms. *Proceedings of the VLDB Endowment* 9, 2 (2015), 36–47. https://doi.org/10.14778/2850578.2850579

[4] Fabian Biester, Mohamed Abdelaal, and Daniel Del Gaudio. 2025. LLM-Clean: Context-Aware Tabular Data Cleaning via LLM-Generated OFDs. In *New Trends in Database and Information Systems (Communications in Computer and Information Science)*, Joe Tekli, Johann Gamper, Richard Chbeir, Yannis Manolopoulos, Salma Sassi, Mirjana Ivanovic, Genoveva Vargas-Solar, and Ester Zumpano (Eds.), Vol. 2186. Springer Nature Switzerland, Cham, 68–78. https://doi.org/10.1007/978-3-031-70421-5_7

[5] Jan-Micha Bodensohn, Ulf Brackmann, Liane Vogel, Anupam Sanghi, and Carsten Binnig. 2025. Unveiling Challenges for LLMs in Enterprise Data Engineering. *Proceedings of the VLDB Endowment* 19, 2 (2025), 196–209. https://doi.org/10.14778/3773749.3773758

[6] Sebastian Bordt, Harsha Nori, Vanessa Rodrigues, Besmira Nushi, and Rich Caruana. 2024. Elephants Never Forget: Memorization and Learning of Tabular Data in Large Language Models. In *First Conference on Language Modeling*. https://openreview.net/forum?id=HLoWN6m4fS

[7] Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramèr, and Chiyuan Zhang. 2023. Quantifying Memorization Across Neural Language Models. In *The Eleventh International Conference on Learning Representations*. https://openreview.net/forum?id=TatRHT_1cK

[8] Nicholas Carlini, Florian Tramèr, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Úlfar Erlingsson, Alina Oprea, and Colin Raffel. 2021. Extracting Training Data from Large Language Models. In *30th USENIX Security Symposium (USENIX Security 21)*. USENIX Association, 2633–2650. https://www.usenix.org/conference/usenixsecurity21/presentation/carlini-extracting

[9] Zui Chen, Lei Cao, Sam Madden, Tim Kraska, Zeyuan Shang, Ju Fan, Nan Tang, Zihui Gu, Chunwei Liu, and Michael Cafarella. 2024. SEED: Domain-Specific Data Curation With Large Language Models. arXiv:2310.00749 [cs.DB] https://arxiv.org/abs/2310.00749

[10] Xu Chu, Ihab F. Ilyas, Sanjay Krishnan, and Jiannan Wang. 2016. Data Cleaning: Overview and Emerging Challenges. In *Proceedings of the 2016 International Conference on Management of Data*. ACM, 2201–2206. https://doi.org/10.1145/2882903.2912574

[11] Xu Chu, Ihab F. Ilyas, and Paolo Papotti. 2013. Holistic data cleaning: Putting violations into context. In *2013 IEEE 29th International Conference on Data Engineering (ICDE)*. IEEE, 458–469. https://doi.org/10.1109/icde.2013.6544847

[12] Xu Chu, John Morcos, Ihab F. Ilyas, Mourad Ouzzani, Paolo Papotti, Nan Tang, and Yin Ye. 2015. KATARA: A Data Cleaning System Powered by Knowledge Bases and Crowdsourcing. In *Proceedings of the 2015 ACM SIGMOD International Conference on Management of Data*. 1247–1261. https://doi.org/10.1145/2723372.2749431

[13] Michele Dallachiesa, Amr Ebaid, Ahmed Eldawy, Ahmed Elmagarmid, Ihab F. Ilyas, Mourad Ouzzani, and Nan Tang. 2013. NADEEF: A commodity data cleaning system. In *Proceedings of the 2013 ACM SIGMOD International Conference on Management of Data*. ACM, 541–552. https://doi.org/10.1145/2463676.2465327

[14] DeepSeek-AI. 2026. DeepSeek-V4: Towards Highly Efficient Million-Token Context Intelligence. arXiv:2606.19348 [cs.CL] https://arxiv.org/abs/2606.19348

[15] Çağatay Demiralp, Fabian Wenz, Peter Baile Chen, Moe Kayali, Nesime Tatbul, and Michael Stonebraker. 2024. Making LLMs Work for Enterprise Data Tasks. arXiv:2407.20256 Poster at North East Database Day 2024.

[16] Bradley Efron and Robert J. Tibshirani. 1994. *An Introduction to the Bootstrap* (1 ed.). Chapman and Hall/CRC. https://www.routledge.com/An-Introduction-to-the-Bootstrap/Efron-Tibshirani/p/book/9780412042317

[17] Raul Castro Fernandez, Aaron J. Elmore, Michael J. Franklin, Sanjay Krishnan, and Chenhao Tan. 2023. How Large Language Models Will Disrupt Data Management. *Proceedings of the VLDB Endowment* 16, 11 (2023), 3302–3309. https://doi.org/10.14778/3611479.3611527

[18] Shahriar Golchin and Mihai Surdeanu. 2024. Time Travel in LLMs: Tracing Data Contamination in Large Language Models. In *International Conference on Learning Representations*, B. Kim, Y. Yue, S. Chaudhuri, K. Fragkiadaki, M. Khan, and Y. Sun (Eds.), Vol. 2024. 43008–43029. https://proceedings.iclr.cc/paper_files/paper/2024/file/bc39a59c49b731c51398ad6b12f301d3-Paper-Conference.pdf

[19] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. 2024. The Llama 3 Herd of Models. arXiv:2407.21783 https://arxiv.org/abs/2407.21783

[20] Alireza Heidari, Joshua McGrath, Ihab F. Ilyas, and Theodoros Rekatsinas. 2019. HoloDetect: Few-Shot Learning for Error Detection. In *Proceedings of the 2019 International Conference on Management of Data*. ACM, 829–846. https://doi.org/10.1145/3299869.3319888

[21] Sture Holm. 1979. A Simple Sequentially Rejective Multiple Test Procedure. *Scandinavian Journal of Statistics* 6, 2 (1979), 65–70. https://www.jstor.org/stable/4615733 JSTOR 4615733.

[22] Ihab F. Ilyas and Xu Chu. 2019. *Data Cleaning*. ACM. https://doi.org/10.1145/3310205

[23] Alon Jacovi, Avi Caciularu, Omer Goldman, and Yoav Goldberg. 2023. Stop Uploading Test Data in Plain Text: Practical Strategies for Mitigating Data Contamination by Evaluation Benchmarks. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. 5075–5084. https://doi.org/10.18653/v1/2023.emnlp-main.308

[24] Nikhil Kandpal, Haikang Deng, Adam Roberts, Eric Wallace, and Colin Raffel. 2023. Large language models struggle to learn long-tail knowledge. In *Proceedings of the 40th International Conference on Machine Learning (Proceedings of Machine Learning Research)*, Vol. 202. PMLR, 15696–15707. https://proceedings.mlr.press/v202/kandpal23a.html

[25] Moe Kayali, Fabian Wenz, Nesime Tatbul, and Çağatay Demiralp. 2025. Mind the Data Gap: Bridging LLMs to Enterprise Data Integration. In *Proceedings of the Conference on Innovative Data Systems Research (CIDR '25)*. Amsterdam, Netherlands. https://www.vldb.org/cidrdb/papers/2025/p34-kayali.pdf

[26] Sanjay Krishnan, Jiannan Wang, Eugene Wu, Michael J. Franklin, and Ken Goldberg. 2016. ActiveClean: Interactive data cleaning for statistical modeling. *Proceedings of the VLDB Endowment* 9, 12 (2016), 948–959. https://doi.org/10.14778/2994509.2994514

[27] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. 2023. Efficient Memory Management for Large Language Model Serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles*. ACM, 611–626. https://doi.org/10.1145/3600006.3613165

[28] Peng Li, Yeye He, Dror Yashar, Weiwei Cui, Song Ge, Haidong Zhang, Danielle Rifinski Fainman, Dongmei Zhang, and Surajit Chaudhuri. 2024. Table-GPT: Table Fine-tuned GPT for Diverse Table Tasks. *Proceedings of the ACM on Management of Data* 2, 3, Article 176 (2024), 28 pages. https://doi.org/10.1145/3654979

[29] Peng Li, Xi Rao, Jennifer Blase, Yue Zhang, Xu Chu, and Ce Zhang. 2021. CleanML: A Study for Evaluating the Impact of Data Cleaning on ML Classification Tasks. In *2021 IEEE 37th International Conference on Data Engineering (ICDE)*. IEEE, 13–24. https://doi.org/10.1109/icde51399.2021.00009

[30] Jiacheng Liu, Sewon Min, Luke Zettlemoyer, Yejin Choi, and Hannaneh Hajishirzi. 2024. Infini-gram: Scaling Unbounded n-gram Language Models to a Trillion Tokens. In *First Conference on Language Modeling*. https://openreview.net/forum?id=u2vAyMeLMm

[31] Shayne Longpre, Kartik Perisetla, Anthony Chen, Nikhil Ramesh, Chris DuBois, and Sameer Singh. 2021. Entity-Based Knowledge Conflicts in Question Answering. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. 7052–7063. https://doi.org/10.18653/v1/2021.emnlp-main.565

[32] Mohammad Mahdavi and Ziawasch Abedjan. 2020. Baran: Effective Error Correction via a Unified Context Representation and Transfer Learning. *Proceedings of the VLDB Endowment* 13, 11 (2020), 1948–1961. https://doi.org/10.14778/3407790.3407801

[33] Mohammad Mahdavi, Ziawasch Abedjan, Raul Castro Fernandez, Samuel Madden, Mourad Ouzzani, Michael Stonebraker, and Nan Tang. 2019. Raha: A Configuration-Free Error Detection System. In *Proceedings of the 2019 International Conference on Management of Data*. ACM, 865–882. https://doi.org/10.1145/3299869.3324956

[34] Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. When Not to Trust Language Models: Investigating Effectiveness of Parametric and Non-Parametric Memories. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 9802–9822. https://doi.org/10.18653/v1/2023.acl-long.546

[35] R. Thomas McCoy, Shunyu Yao, Dan Friedman, Mathew D. Hardy, and Thomas L. Griffiths. 2024. Embers of autoregression show how large language models are shaped by the problem they are trained to solve. *Proceedings of the National Academy of Sciences* 121, 41 (2024), e2322420121. https://doi.org/10.1073/pnas.2322420121

[36] Iman Mirzadeh, Keivan Alizadeh-Vahid, Hooman Shahrokhi, Oncel Tuzel, Samy Bengio, and Mehrdad Farajtabar. 2025. GSM-Symbolic: Understanding the Limitations of Mathematical Reasoning in Large Language Models. In *International Conference on Learning Representations*, Y. Yue, A. Garg, N. Peng, F. Sha, and R. Yu (Eds.), Vol. 2025. 94743–94765. https://proceedings.iclr.cc/paper_files/paper/2025/file/ec2e7a896f8250986b3907f57621ce94-Paper-Conference.pdf

[37] Avanika Narayan, Ines Chami, Laurel Orr, and Christopher Ré. 2022. Can Foundation Models Wrangle Your Data? *Proceedings of the VLDB Endowment* 16, 4 (2022), 738–746. https://doi.org/10.14778/3574245.3574258

[38] Ella Neeman, Roee Aharoni, Or Honovich, Leshem Choshen, Idan Szpektor, and Omri Abend. 2023. DisentQA: Disentangling Parametric and Contextual Knowledge with Counterfactual Question Answering. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 10056–10070. https://doi.org/10.18653/v1/2023.acl-long.559

[39] Felix Neutatz, Mohammad Mahdavi, and Ziawasch Abedjan. 2019. ED2: A Case for Active Learning in Error Detection. In *Proceedings of the 28th ACM International Conference on Information and Knowledge Management*. 2249–2252. https://doi.org/10.1145/3357384.3358129

[40] Wei Ni, Xiaoye Miao, Xiangyu Zhao, Yangyang Wu, Shuwei Liang, and Jianwei Yin. 2024. Automatic Data Repair: Are We Ready to Deploy? *Proceedings of the VLDB Endowment* 17, 10 (2024), 2617–2630. https://doi.org/10.14778/3675034.3675051

[41] Wei Ni, Kaihang Zhang, Xiaoye Miao, Xiangyu Zhao, Yangyang Wu, Yaoshu Wang, and Jianwei Yin. 2025. ZeroED: Hybrid Zero-Shot Error Detection with Large Language Model Reasoning. In *2025 IEEE 41st International Conference on Data Engineering (ICDE)*. 3126–3139. https://doi.org/10.1109/ICDE65448.2025.00234

[42] Wei Ni, Kaihang Zhang, Xiaoye Miao, Xiangyu Zhao, Yangyang Wu, and Jianwei Yin. 2024. IterClean: An Iterative Data Cleaning Framework with Large Language Models. In *Proceedings of ACM Turing Award Celebration Conference - CHINA 2024*. Association for Computing Machinery, 100–105. https://doi.org/10.1145/3674399.3674436

[43] Curtis G. Northcutt, Anish Athalye, and Jonas Mueller. 2021. Pervasive Label Errors in Test Sets Destabilize Machine Learning Benchmarks. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, J. Vanschoren and S. Yeung (Eds.), Vol. 1. https://datasets-benchmarks-proceedings.neurips.cc/paper_files/paper/2021/file/f2217062e9a397a1dca429e7d70bc6ca-Paper-round1.pdf

[44] OpenAI. 2026. GPT-6.1 Sol. https://openai.com/index/introducing-gpt-6-1-sol/ Accessed: 2026-10-03.

[45] Yonatan Oren, Nicole Meister, Niladri Chatterji, Faisal Ladhak, and Tatsunori Hashimoto. 2024. Proving Test Set Contamination in Black-Box Language Models. In *International Conference on Learning Representations*, B. Kim, Y. Yue, S. Chaudhuri, K. Fragkiadaki, M. Khan, and Y. Sun (Eds.), Vol. 2024. 16354–16372. https://proceedings.iclr.cc/paper_files/paper/2024/file/46e624c244cff669223d488defd4e835-Paper-Conference.pdf

[46] Ralph Peeters, Aaron Steiner, and Christian Bizer. 2025. Entity Matching using Large Language Models. In *Proceedings of the 28th International Conference on Extending Database Technology*. OpenProceedings.org, 529–541. https://doi.org/10.48786/edbt.2025.42

[47] Fabio Petroni, Tim Rocktäschel, Sebastian Riedel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, and Alexander Miller. 2019. Language Models as Knowledge Bases?. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. 2463–2473. https://doi.org/10.18653/v1/d19-1250

[48] Danrui Qi, Zhengjie Miao, and Jiannan Wang. 2024. CleanAgent: Automating Data Standardization with LLM-based Agents. arXiv:2403.08291 https://arxiv.org/abs/2403.08291 Revised June 2025; arXiv:2403.08291v4.

[49] Qwen, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, et al. 2024. Qwen2.5 Technical Report. arXiv:2412.15115 https://arxiv.org/abs/2412.15115

[50] Qwen Team. 2026. Qwen3.8-Max: A New Bar for Coding and Cowork. https://qwen.ai/blog?id=qwen3.8

[51] Erhard Rahm and Hong Hai Do. 2000. Data cleaning: Problems and current approaches. *IEEE Data Eng. Bull.* 23, 4 (2000), 3–13. https://dbs.uni-leipzig.de/research/publications/data-cleaning-problems-and-current-approaches

[52] Theodoros Rekatsinas, Xu Chu, Ihab F. Ilyas, and Christopher Ré. 2017. HoloClean: Holistic data repairs with probabilistic inference. *Proceedings of the VLDB Endowment* 10, 11 (2017), 1190–1201. https://doi.org/10.14778/3137628.3137631

[53] Jayanta Sadhu, Sayem Shahad, and Kenneth Marino. 2026. DERELAB: Probing Defeasible Reasoning and Confirmation Bias in LLMs with a Generative Benchmark. https://doi.org/10.48550/arXiv.2608.30413 arXiv:2608.30413 Accepted at EMNLP 2026.

[54] Oscar Sainz, Jon Ander Campos, Iker García-Ferrero, Julen Etxaniz, Oier Lopez de Lacalle, and Eneko Agirre. 2023. NLP Evaluation in trouble: On the Need to Measure LLM Data Contamination for each Benchmark. In *Findings of the Association for Computational Linguistics: EMNLP 2023*. 10776–10787. https://doi.org/10.18653/v1/2023.findings-emnlp.722

[55] Abulhair Saparov and He He. 2023. Language Models Are Greedy Reasoners: A Systematic Formal Analysis of Chain-of-Thought. In *The Eleventh International Conference on Learning Representations*. https://openreview.net/forum?id=qFVVBzXxR2V

[56] Weijia Shi, Anirudh Ajith, Mengzhou Xia, Yangsibo Huang, Daogao Liu, Terra Blevins, Danqi Chen, and Luke Zettlemoyer. 2024. Detecting Pretraining Data from Large Language Models. In *International Conference on Learning Representations*, Vol. 2024. 51826–51843. https://proceedings.iclr.cc/paper_files/paper/2024/file/e32ad85fa27be4a9868d55703f01323e-Paper-Conference.pdf

[57] Matteo Silvestri, Fabiano Veglianti, Flavio Giorgi, Fabrizio Silvestri, and Gabriele Tolomei. 2026. When Large Language Models Know the Table: A Framework for Assessing Data Contamination in Tabular Datasets. In *Conference on Language Modeling (COLM)*. arXiv:2510.20351 https://arxiv.org/abs/2510.20351 arXiv:2510.20351v4.

[58] Jianhong Tu, Ju Fan, Nan Tang, Peng Wang, Guoliang Li, Xiaoyong Du, Xiaofeng Jia, and Song Gao. 2023. Unicorn: A Unified Multi-tasking Model for Supporting Matching Tasks in Data Integration. *Proceedings of the ACM on Management of Data* 1, 1, Article 84 (2023), 26 pages. https://doi.org/10.1145/3588938

[59] Denny Vrandečić and Markus Krötzsch. 2014. Wikidata: A free collaborative knowledgebase. *Commun. ACM* 57, 10 (2014), 78–85. https://doi.org/10.1145/2629489

[60] Evan Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, Nathan Lambert, et al. 2025. 2 OLMo 2 Furious (COLM's Version). In *Second Conference on Language Modeling*. https://openreview.net/forum?id=2ezugTT9kU

[61] Jiannan Wang and Nan Tang. 2014. Towards dependable data repairing with fixing rules. In *Proceedings of the 2014 ACM SIGMOD International Conference on Management of Data*. ACM, 457–468. https://doi.org/10.1145/2588555.2610494

[62] Xinzhi Wang and Chunwei Liu. 2026. The Table Says Otherwise: Testing LLMs with Counterfactual Relational Data. arXiv:2606.23667 https://arxiv.org/abs/2606.23667

[63] Yibo Wang, Riteng Zhang, Yinghao He, Yongye Su, Bharat Bhargava, and Chunwei Liu. 2026. TabClean: Reusable LLM-Synthesized Programs for Tabular Data Cleaning. arXiv:2606.25388

[64] Colin White, Samuel Dooley, Manley Roberts, Arka Pal, Benjamin Feuer, Siddhartha Jain, Ravid Shwartz-Ziv, Neel Jain, Khalid Saifullah, Sreemanti Dey, Shubh-Agrawal, Sandeep Singh Sandha, Siddartha Naidu, Chinmay Hegde, Yann LeCun, Tom Goldstein, Willie Neiswanger, and Micah Goldblum. 2025. LiveBench: A Challenging, Contamination-Limited LLM Benchmark. In *International Conference on Learning Representations*, Vol. 2025. 91595–91631. https://proceedings.iclr.cc/paper_files/paper/2025/hash/e4a46394ba5378b3f9a186a5b4c650d1-Abstract-Conference.html

[65] Zhaofeng Wu, Linlu Qiu, Alexis Ross, Ekin Akyürek, Boyuan Chen, Bailin Wang, Najoung Kim, Jacob Andreas, and Yoon Kim. 2024. Reasoning or Reciting? Exploring the Capabilities and Limitations of Language Models Through Counterfactual Tasks. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. 1819–1862. https://doi.org/10.18653/v1/2024.naacl-long.102

[66] Jian Xie, Kai Zhang, Jiangjie Chen, Renze Lou, and Yu Su. 2024. Adaptive Chameleon or Stubborn Sloth: Revealing the Behavior of Large Language Models in Knowledge Conflicts. In *International Conference on Learning Representations*, Vol. 2024. 35623–35646. https://proceedings.iclr.cc/paper_files/paper/2024/file/99261adc8a6356b38bcf999bba9a26dc-Paper-Conference.pdf

[67] Cheng Xu, Shuhao Guan, Derek Greene, and M-Tahar Kechadi. 2024. Benchmark Data Contamination of Large Language Models: A Survey. https://doi.org/10.48550/arXiv.2406.04244 arXiv:2406.04244

[68] Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. 2024. Knowledge Conflicts for LLMs: A Survey. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. 8541–8565. https://doi.org/10.18653/v1/2024.emnlp-main.486

[69] Z.ai. 2026. GLM-5.3: Frontier Coding with Emergent Cyber Capabilities. https://z.ai/blog/glm-5.3 Published August 14, 2026; accessed October 20, 2026.

[70] Haochen Zhang, Yuyang Dong, Chuan Xiao, and Masafumi Oyamada. 2024. Jellyfish: Instruction-Tuning Local Large Language Models for Data Preprocessing. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. 8754–8782. https://doi.org/10.18653/v1/2024.emnlp-main.497

[71] Shuo Zhang, Zezhou Huang, and Eugene Wu. 2025. Data cleaning using large language models. In *2025 IEEE 41st International Conference on Data Engineering Workshops (ICDEW)*. IEEE, 28–32. https://doi.org/10.1109/ICDEW67478.2025.00008

[72] Mark Ziemann, Yotam Eren, and Assam El-Osta. 2016. Gene name errors are widespread in the scientific literature. *Genome Biology* 17, 1 (2016), 177. https://doi.org/10.1186/s13059-016-1044-7